

BUSN 32120 · FINAL PROJECT

Which economic environment is better for a beginner investor?

How Fed policy, inflation, recessions, and S&P 500 returns shape the choice between cash and stocks

Data FRED macro · S&P 500 · NBER recessions | **Tools** Python · SQL

Team 5 · Spring 2026



A high savings rate is not automatically a good return

Our question

When are cash-like returns attractive, and when does the stock market look more favorable?

The nominal rate alone does not tell the full story. The key concept is the real rate: the nominal rate minus inflation.

WHO IT'S FOR

Beginner investors and finance students deciding whether to hold more cash or invest in the S&P 500

WHY IT MATTERS

Real rate = nominal rate – inflation

If inflation is higher than the cash return, then the investor is still **losing purchasing power** — even when the headline rate looks good.

Three datasets, two languages, one aligned sample

FRED MACRO

Fed Funds Rate, CPI inflation, unemployment

S&P 500

Monthly stock returns (level → within-month return)

RECESSION INDICATOR

Compare expansions and recessions

PYTHON

cleaning · feature engineering · EDA · charts · 2 models

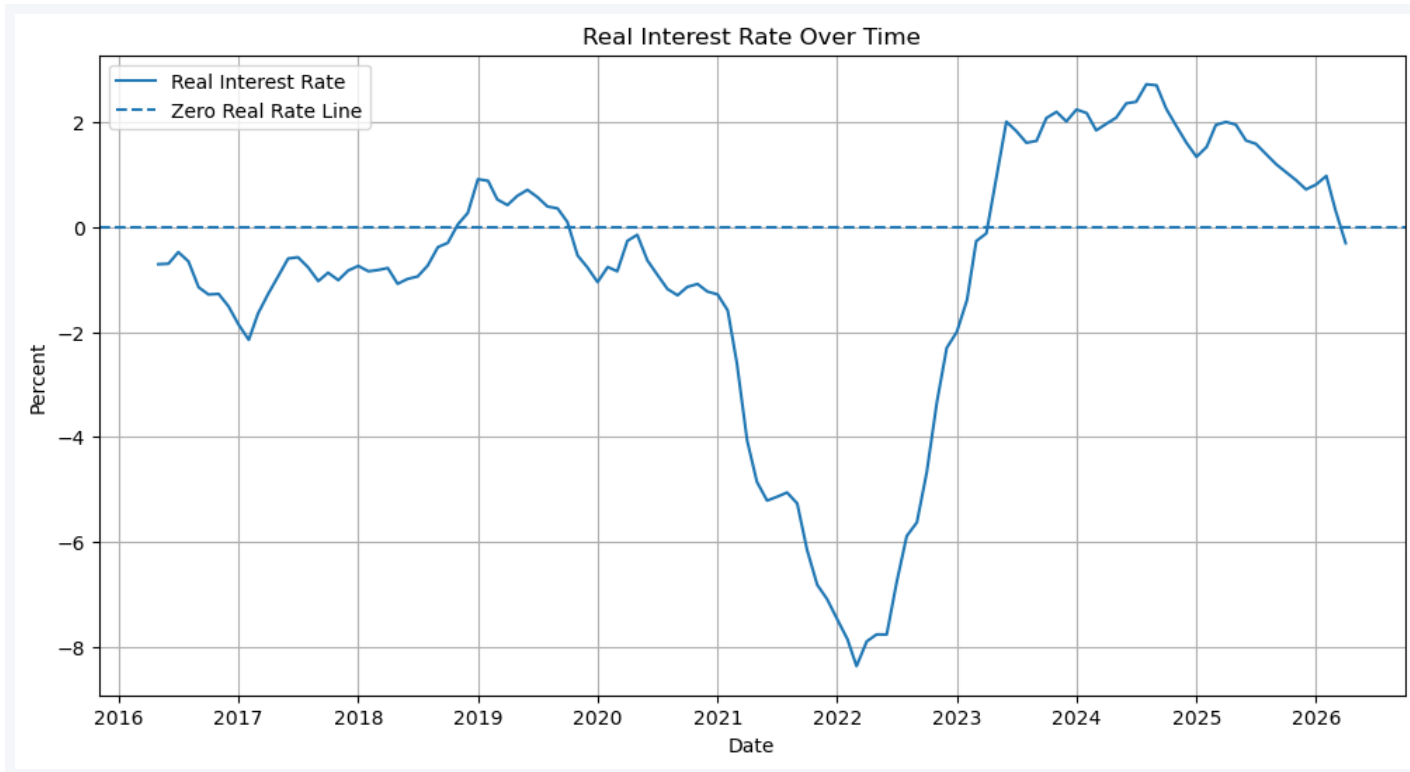
SQL

joins · window functions · group by · subqueries

SCOPE:

The whole analysis uses one aligned sample: **the most recent 119 complete months, 2016–2026**. We trim to where **all FRED series overlap** — the FRED S&P 500 series only goes back to 2015, so we keep macro and market on the same 119-month window.

Real rates matter much more than nominal rates



From our notebook — Chart 2: real interest rate over time (dashed = zero line)

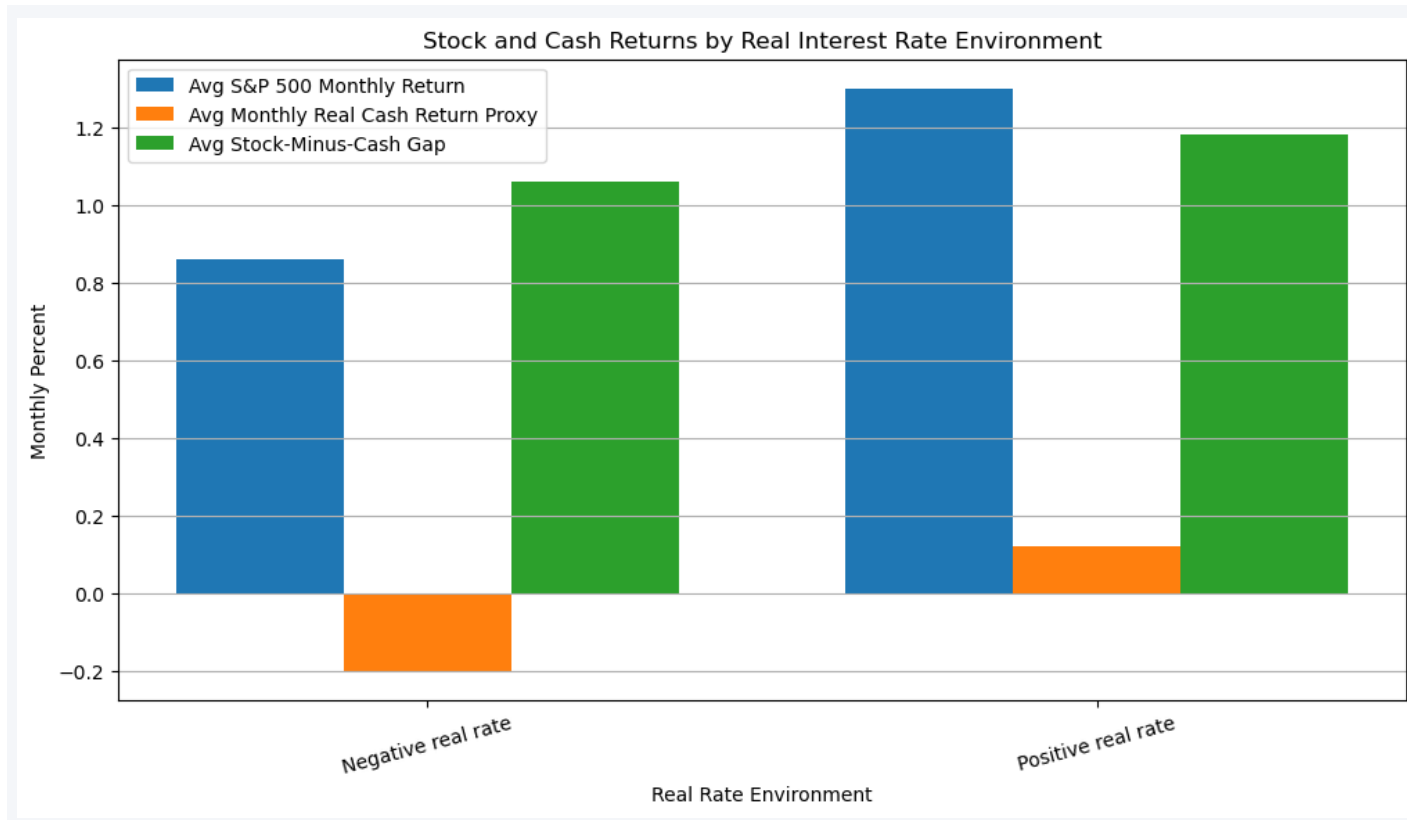
61%

of months (73 of 119) had a negative real rate —
cash was losing purchasing power in real terms

So what?

Real rates were deeply negative around 2021–
2022 when inflation spiked, then turned positive
after 2023 as the Fed raised rates.

Cash becomes much more defensible when the real rate is positive



Reading the chart

When real rates are negative, the cash proxy loses about 0.20%/mo. When real rates are positive, cash earns about 0.12%/mo — it is actually protecting purchasing power.

In our available S&P 500 sample, stocks are positive in both environments and higher than the cash proxy — but the S&P sample is shorter, so this evidence cannot be used as a rule.

Takeaway: whether cash is good or bad depends on the real-rate environment.

Macro is weak for prediction, but is a good indicator of the environment

MODEL 1 · LINEAR REGRESSION

Explain monthly S&P 500 returns from macro variables

$$R^2 \approx -0.35$$

A test R^2 of about -0.35 means the model does **worse than predicting the average return**. Monthly returns are noisy and macro isn't a good short-term trading signal. The R^2 value is showing the limit of macro for forecasting monthly equity returns.

MODEL 2 · LOGISTIC REGRESSION

Classify high-inflation months (YoY inflation $> 3\%$)

ROC AUC 0.91

Accuracy 0.67 · Precision 0.50 · Recall 1.00 · F1 0.67

Falling unemployment and increasing rate are useful signals for high-inflation environments, which lead to tighter labor markets.

Real beats nominal; macro is context, not a trading signal; know your sample

Real beats nominal

Cash is truly attractive when the return clears inflation. In this sample about three in five months had a negative real rate.

Context, not signal

Macro indicators help investors understand the environment, but they don't reliably predict monthly S&P 500 returns.

Know your sample

This run uses one aligned 119-month window (2016–2026), because the FRED S&P 500 series only starts in 2015.

NEXT Add T-bill yields, bond ETF returns, and consumer sentiment; backfill a true long-history S&P 500 source, which would make the market return analysis stronger.